Electrical Influences and Speed of Chemical Change in the Brain

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CHEMICAL CHANGE IN THE BRAIN is large in extent, is intricate, rapid, and closely connected with electrical phenomena. In man under basal metabolic conditions, some 20 per cent of the oxygen taken up by the body in respiration is consumed in the brain. In reviewing the speed of chemical change in the brain one is thus concerned with an appreciable proportion of the energy metabolism of the body: much of the present account assesses the respiration, carbohydrate metabolism, and energy-rich intermediates of the brain and of parts of the brain; it is especially concerned with the manner in which these metabolic characteristics are affected by experimentally applied electrical influences. Other aspects of the speed of chemical change in the brain have been described elsewhere (73).

RESPIRATION

In Vivo. Much quantitative information on how the level of cerebral metabolism varies with the functional state of the brain has come in the last decade from analysis of arterial and cerebral venous blood (51, 52, 73, 95). This has shown that a normal rate of cerebral oxygen consumption in man or the monkey, of some 90 μM/gm of brain/hr. (3.3–3.8 ml O_2/100 gm/min.) can be reduced to about 55 μM/gm/hr. in anesthesia and can be increased by convulsive agents. The highest rate recorded was of 140 μM/gm/hr. during convulsions induced by pentamethylenetetrazole. Respiration probably increased very promptly, for potentiometric measurements showed a rapid fall in oxygen tension in the cerebral cortex at or before the beginning of overt convulsions.

Cerebral respiration does not appear to have been measured during excitation induced electrically. This is probably because the period of excitation is short and measurement ordinarily requires observations over a period of 10 minutes. Consequently the post-convulsive state is more accessible to study, and in this phase, in schizophrenic patients who otherwise were of normal cerebral respiratory rate, electroshock transitorily lowered this rate by 17 per cent (53). Probably, however, during the actual electrical stimulation of the brain, cerebral respiration rises. This is judged from other chemical similarities between excitation induced electrically and by pentamethylenetetrazol (see below), from potentiometric observations (23) and also from the prolonging of electrically-induced convulsions by increased oxygen tension (94).

Perfused and Isolated Brain. The cat brain perfused with defibrinated blood consumed oxygen at 90–130 μM/gm/hr. (20, 95), rates close to the values noted above in intact mammals. Perfusion with simpler solutions has also been carried out (see section on GLUCOSE AND GLYCOLYSIS). Metabolites which in perfusion reach the brain through the blood capillaries have not been adequately provided to the entire mammalian brain by other means. Of the metabolites normally coming from the blood,

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oxygen is probably the first limiting factor in short-term experiments, for the brain contains appreciable oxidizable substrates including glucose and glycogen. To portions of mammalian tissue less than 0.4 millimeters in thickness, and at 37°, oxygen with other substances can be provided by diffusion from a bathing solution, as is described below. The brain in many lower vertebrates is, however, of smaller size than in mammals and also is of lower respiratory rate, especially at the lower temperatures normal to cold-blooded animals. This has enabled respiratory observations to be made with the intact brain and spinal cord of the frog and of small fishes. Moreover, such preparations contain sufficient endogenous substrates for experiments of an hour or more. Thus the addition of fluids can be minimized, and the frog brain studied under simple conditions comparable to those employed with peripheral nerve.

Studied in this way mainly in capillary respirometers, frog spinal cord consumed at 18°C or 12–15 μM O_2/gm/hr., frog brain 18 μmoles/g/hr., and the brain of different fishes 8–10 μM/gm/hr. (106). As determination of respiratory rate by arterial-venous difference in the living animal is not available in these species, it is important to note that in some experiments connections from the frog spinal cord to the leg were left intact, so that the excitability of the cord could be examined; or recording electrodes were placed in the cord or on the brain itself. These showed maintained excitability or spontaneous potential changes in the isolated organs (7, 64, 106). The respiratory rates described are thus likely to be relevant to the functioning organs.

Electrical stimulation. Several groups of experiments under the conditions just described demonstrated that applied electrical pulses increased the respiration of the brain and spinal cord of frogs or fishes (106). By pulses from induction coils, respiration of the brain could be maintained for an hour or so at levels 20–35 per cent above its normal value; it reverted to this level when pulses were stopped. Pulses applied to the frog cord sufficiently to maintain tetanus in the limb for an hour increased the respiration of the cord during this period by 30 to 60 per cent; this also fell to normal when pulses were stopped, and could be increased again by further pulses. When intensity of the pulses (indicated only by position of induction coils) was increased, respiratory response increased by a maximum of 200 per cent of the existing level.

Separated Tissues. Only after many years were experiments comparable to those with the amphibian brain made successfully with parts of the mammalian brain (65). The respiratory rate of the greater part of the mammalian brain requires that for adequate diffusion of oxygen to a piece of tissue, it be not greater than about 0.4 millimeters in thickness. In white matter some 0.8 millimeters may be permissible. Pieces of cerebral cortex adequately oxygenated do not appear to show spontaneous changes in potential as does the frog brain, nor are transmitted electrical disturbances easily observed after applying electrical stimuli to them (80). Also, respiratory rates of cerebral tissues in ordinary glucose salines are lower than observed *in vivo*. For white matter the rates *in vitro* are about 30 μM O_2/gm/hr. and for grey matter about 70; the divergence from *in vivo* values was accentuated in the opinion of earlier workers by erroneously high values *in vivo* (95) and, often, low values *in vitro*. The separated tissues may therefore have been judged unsuitable for study of possible metabolic response to applied pulses. Further, mammalian cerebral tissues lose the greater part of their energy-rich metabolites on which electrical activity is likely to depend, during their preparation for typical experiments *in vitro* (see below).

It was, however, discovered that the separated tissue, respiring in ordinary

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salines with suitable substrates, could be caused to resynthesize and maintain a pattern of energy-rich metabolites similar to that *in vivo* (68, 76). Stimulating agents including electrical pulses deplete these substances *in vivo*; it was therefore relevant to examine their effects *in vitro*. Applied electrical pulses were then found to alter not only the tissue's composition but also its respiratory rate. Respiration was accelerated by the pulses and indeed found to rise to the higher levels typical of the brain *in vivo*.

Electrical pulses are not the only agent capable of increasing the respiratory rate of cerebral tissues; notably, 2:4-dinitrophenols and 30 mM potassium salts can do so. These are not however a normal part of the environment of cerebral tissues; their actions are assessed below.

Experimental arrangements and apparatus for applying pulses. For observing respiratory response to applied pulses, satisfactory metabolic conditions must be maintained or restored to the separated cerebral tissue. For these purposes, and also for measuring gas exchange which indicates the tissue's condition and respiratory response, apparatus has been employed which approaches the conventional types in biochemical use. Conceivably very different apparatus such as certain types developed for respiratory observations with peripheral nerve and ganglia (21, 61) would also be applicable. Most observations with cerebral tissues have been made with constant-volume manometers and conical vessels of some 15 milliliters, taking some 50 milligrams of tissue in about 3 milliliters of fluid (65, 9); in modified apparatus these quantities have been decreased to $\frac{1}{5}$ or increased 3-fold. The fluids may be based on those of Krebs (60, 65, 92).

When labile metabolites are resynthesized, much of the excess fluid may be removed or replaced by oils, and metabolic response remains (92). Under these circumstances evidence has been sought for transmitted electrical effects, so far unsuccessfully. Consequently methods have been developed for observing metabolic response independently of electrical measurements, and this has resulted in experimental arrangements characteristically different from current practice with other parts of the nervous system. Thus, the presence of much fluid has been accepted as a metabolic convenience though it would make observation of electrical response difficult, and currents have been applied sufficient to give potential gradients adequate for excitation, throughout a considerable bulk of fluid. Pulses by charge and discharge of condensers are more suitable for this purpose than those from commercial square wave generators; consequently much use has been made of condenser pulses and sine-wave alternating currents. Under most experimental circumstances with cell-containing tissues and with electrodes of silver, gold or molybdenum, no difficulties arise from application of brief pulses of up to 20 V in such systems (65, 9); complications under other circumstances are noted below.

Metabolic requirements for respiratory response. Respiratory response by guinea pig or rat cerebral cortex changes little when the bicarbonate, phosphate, chloride, sulphate, potassium and magnesium ions of physiological salines are replaced by other ions (30, 103). Respiration itself increases, and respiratory response decreases when potassium salts are increased in concentration or calcium salts decreased; added ammonium salts also have this effect (30, 105). The tonicity of the medium as a whole can be increased or decreased some 30 per cent without great effect on respiration or respiratory response, and also may have glycylglycine or sodium nitrate as its major component.

Sodium salts show an interestingly specific relationship to the stimulation. When decreased from the normal 130 mM to 15 mM, normal respiration was lowered only a

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little but respiratory response to pulses was almost absent. This was found with pulses of different characteristics which in ordinary media gave partial or maximum response, or were in excess of those needed to give maximum response (30); the phenomenon is thus comparable to the loss of excitability seen in muscle or the squid axon in sodium deficiency (45, 88).

Without added oxidizable substrate cerebral tissues respire, but pulses do not increase their respiratory rate. The rate is initially lower than with glucose and falls with time; species variations have been noted in this respect (102) and pulses tend to accelerate the fall (79). Respiratory rate and response in ordinary salines with glucose at 10 mM remains unaffected if glucose is altered between 25 and 2.5 mM, but below 2.5 mM respiratory response falls to a much greater extent than does respiratory rate without pulses. Pulses thus make respiration in the separated tissue more akin to that in the whole animal in its sensitivity to hypoglycaemia (69). Maximum respiratory response is yielded when glucose is replaced by pyruvates at about 1 mM or greater, and by lactates or fructose when these are present at markedly higher levels of some 10 mM (73). Most other substrates are however less effective. Little or no response is yielded by cerebral cortex of rat or guinea pig with succinate, α-ketoglutarate, citrate, or fumarate though respiration without pulses is comparable to that with glucose (57). Response with glutamic acid as substrate, small or absent in these species, is, however, appreciable in human cerebral tissues (70). Other substrates here are again without effect. In many of these instances, absence of respiratory response to pulses has been correlated with lack of ability to maintain the labile phosphates of the tissue.

Electrical requirements for respiratory response. TIME FACTORS. In the several parts of the brain and the several species examined, a sustained respiratory response requires a maintained series of pulses. It follows within 30 seconds of the beginning of their application and ceases within a minute of their stopping; these times represent the limit of measurements made and response may actually be much prompter (65, 72). When a series of pulses at 100 per second, themselves giving maximum respiratory response from cerebral cortical tissue, is delivered intermittently, response can nevertheless remain high. Thus, when pulses were delivered for 0.2 second each second respiratory rate fell only 14 per cent from its maximum; but when applied for 2 seconds each 10 seconds, response was little more than $\frac{1}{5}$ of normal. These and other observations can be interpreted as indicating a brief persistence of respiratory effect after pulses cease in each cycle of events. Other interpretations are also possible, but assuming persistence, the effect of pulses was calculated to decay with a time-constant of about 3 seconds. This tentative value for cerebral cortical tissues compares with directly observed values of 10–20 minutes in peripheral nerve and 4 minutes in ganglia (61).

SPATIAL SPREAD. The effect of pulses is limited in space. Electrodes placed 2 millimeters apart at one end of a strip of cerebral cortex cut parallel to its outer surface and a centimeter or so long, have much smaller respiratory effect than electrodes of alternating polarity placed at 2-millimeter intervals all along the slice. Maximum metabolic response can, however, follow when fragments of tissue, for example of some 0.1 to 10 milligrams in weight, are floating freely in salines in which potential gradients are established by pulses applied to electrodes some millimeters away, provided the potential applied to the electrodes is correspondingly increased (9, 66, 72). The potential gradients established by pulses between electrodes at different distances apart have been measured. In apparatus convenient metabolically, they are

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not uniform but such data enables approximate comparisons to be made between the gradients and the increase in respiration which they cause. In this way it was found that, for example, a 25 per cent increase in respiration required a peak potential gradient over the greater part of the fluid, of some 0.4 v per millimeter whether the electrodes were 5 or 10 millimeters apart. The actual potential applied was about twice as great in the second case.

FREQUENCY. With pulses of exponential time-voltage relationship and of given voltage and frequency, response increases with increasing frequency of their application. Increase is approximately linear until respiration has increased by some 60 or 80 per cent, after which a maximum about twice the resting level is approached more gradually. Thus, pulses of time-constant 0.4 millisecond giving a peak potential gradient of 1.8 v per millimeter in saline surrounding fragments of guinea pig cerebral cortex, increased respiration by some 25 per cent when applied at 3 per second and 75 per cent when applied at 10 per second; at 100 per second, however, respiration was only a little greater (72).

PULSE DURATION. Respiratory response can be approximately proportional to the time-constant of condenser pulses, applied as just described. Again, the upper limit to the increased respiration is about double the resting level. Thus pulses at 10 per second which gave a respiratory response of 38 per cent when their time-constant was 0.2 millisecond, gave some 75 per cent when, other characteristics being equal, their time-constant was 0.4 millisecond.

It thus occurs that over suitable ranges of values, response is proportional to the product of frequency and time-constant. A plot of respiratory response against this product gave a single straight line with pulses of frequencies of 3 or 10 per second and time-constants up to 0.4 or 1 millisecond. However, pulses at 100 per second were in this sense less effective, and at 1000 per second markedly less so (72).

POTENTIAL. Respiratory response increases with increasing voltage applied, including again a range in which response is nearly directly proportional to applied potential. This may be the center part of a sigmoid curve; at higher voltages a maximum about twice the resting level is again approached but at low voltages the response-voltage curve starts fairly steeply and needs little extrapolation to afford a 'threshold' voltage. Thus condenser pulses which at 2.5 v yielded a 20 per cent (S.D.5) and at 5 v, 64 per cent (S.D.7) response, at 1.5 v yielded 2 per cent (S.D.6; see ref. 72). Similar relationships are seen with sine-wave alternating currents. Response-voltage curves so determined are not suitable for detailed analysis; it will be recalled that potential gradient is not uniform throughout the fluid.

SINE-WAVE ALTERNATING CURRENTS. Here frequency and duration of the voltage change are necessarily altered together. This results, in cerebral tissues as in other excitable systems, in an optimum frequency or range of frequencies at which given responses are obtained at minimum potential (32, 72). This range was found to be about 100 cycles per second with guinea pig cerebral cortex; at 100 cycles per second, response could be obtained when current at a virtual potential of 0.4 v was applied to electrodes 2 millimeters apart. The 'threshold' voltage rose to about 0.7 v at 20 cycles per second and to 1.4 v at 1000 cycles per second. Threshold/frequency relationships established in peripheral nerve (48) possibly held approximately in the cerebral tissue. At frequencies around the optimal, respiration could again be nearly doubled in speed by potentials of some 1.5 v r.m.s.; for excitation with these currents much more electrical energy was expended in the fluid than when the brief pulses of the preceding paragraphs were employed.

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Tissues responding. Respiratory response to pulses applied to cerebral tissues is not uniform throughout an animal's life but greatly increases in the rat and guinea pig at times when other aspects of cerebral functioning are developing (33). The two species contrast interestingly in this respect. In the guinea pig, observations were made with sections of cerebral cortex from animals of different ages from about 30 days gestation. At this age a respiratory rate of 27 μM O_2/gm/hr. was increased only to 32 μM/gm/hr. by pulses which with tissues taken at 35 days gestation caused a 40 per cent increase, and at 57 days, one of 110 per cent (from 36 to 76 μM/gm/hr.). The percentage change in respiration remained at about this level in tissue from newborn and adult animals when, however, the rate in absence of pulses was much higher, about 60 μM/gm/hr. These changes in excitability which occurred in the guinea pig brain prenatally were found to take place in the rate after birth, the most rapid increase in respiratory response occurring between 6 and 15 days of age, while maximum respiratory rate was reached at about 35 days of age. Observations on the electrical activity and excitability of the brain *in situ* show many similarities with those just described in the separated tissue. Electrical activity in the brain of the rat appeared only after birth, when it was at first intermittent, approaching that of the adult at 10–14 days; strychnine did not elicit large potentials until the 4th day (22). In the guinea pig, electrical activity first became regular at 46 days gestation; motor responses to electrical stimulation appeared at 43–46 days (47, 54).

Abnormal cerebral tissues have shown decreased respiratory response to electrical pulses. Cerebral tumors, or areas of gliosis induced by freezing, showed little respiratory response although in some cases their level of respiration without applied pulses was almost normal (33, 71). When different parts of the body were examined under conditions in which cerebral tissues respond, only nervous and muscular tissues showed changes in respiration. Response was notably absent in liver and kidney preparations; respiration of rat diaphragm increased some 60 per cent and that of preparations from the heart also increased (56).

Cell-Free Preparations. *Transition from cell-containing to cell-free preparations.* Effects may first be recounted of subdividing cerebral tissues and examining their respiratory response to pulses in ordinary glucose salines resembling in composition the cerebrospinal fluid (77). Minimal cutting gives a sheet some 20 x 10 x 0.35 millimeters which may be considerably subdivided without greatly affecting either its normal respiration or the increase in respiration caused electrically. Thus the grey matter from rat or guinea pig may be chopped to prisms 0.2 x 0.2 x 2 millimeters with fall of some 5 per cent in these characteristics: the weight of an individual fragment is then about 1/1000 of the sheet and the total superficial area has been increased 4-fold. With 10-fold increase in area (at 0.067 x 0.067 x 2 mm) respiratory response fell about 30 per cent. When ground, respiration was at about half its level in the intact slice and respiratory response was lost. Response thus appears to need cellular structure.

Solutions analogous to extracellular fluids do not, however, represent optimal conditions for studying respiration of tissue elements after loss of cell structure. Experience with cerebral preparations (11, 15, 85) has led to the use of mixtures containing phosphate and magnesium salts, fumarate or malate, pyruvate, diphosphopyridine nucleotide and adenine nucleotides. Under such circumstances the ground tissue can respire at rates of some 120 μM/gm/hr., already approaching those maximal for the cell-containing tissue when maximally increased by electrical pulses or otherwise. Further, these conditions with the ground tissue permit respiration to be coupled

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with the esterification of inorganic phosphate and so to retain as phosphate-bound energy much of the energy derived from respiration. To adequately measure phosphorylation there may, however, also be included in the reaction mixture fluorides and nicotinamide to limit the loss of coenzymes and glucose and further hexokinase to act as phosphate acceptors. Phosphorylation can then occur at some 300 micromoles per gram original tissue per hour.

Electrical pulses applied to ground cerebral tissues in such reaction mixtures again have little if any effect on their respiration, when pulse types are used which are fully effective with cell-containing tissues, and applied in the same apparatus (66, 84). It will be noted, however, that metabolic measurements in such systems are made on a very different footing from those in cell-containing tissues in typical physiological salines. Observations in the cell-free mixtures were made under conditions chosen to exhibit the maximum respiratory capacity of the mixture. If a reaction mixture had been chosen on the basis of allowing cell-containing cerebral tissues to respire at maximal rates, it would contain some 40 mM potassium salts, no calcium salts and also added glutamate, fumarate, or succinate (59, 73). Under these circumstances the slices respire at some 120 μM/gm/hr., they no longer accumulate labile phosphates and their respiration does not increase with applied electrical pulses. It is not surprising, therefore, that no response to pulses was found in the cell-free preparations in the reaction mixtures of the preceding paragraph.

From ground cerebral as other tissues, the component responsible for respiration and phosphorylation has been separated and recognized as the tissue's mitochondria (13, 14). Employing mitochondrial preparations also, little or no change in respiration was found to be associated with the application of electrical pulses adequate for doubling the respiratory rate of cell-containing tissues when examined in the same apparatus (84).

Technical considerations. For experimental details the original papers must be consulted, but certain points are discussed here which have led to electrical influences being reported to increase by about 100 per cent the respiration of suspensions of cerebral tissues and of mitochondria from them (1, 3). These results were reported in an experimental arrangement which, although it was stated to be the same as or to be only slightly modified from that in which the negative results just described were obtained, actually differed from it in the following ways. *a*) Pulses were differently derived and apparently unidirectional and not alternating; electrodes were differently constructed and differently disposed in the fluid. *b*) Electrodes were of different materials. Although initially (3) stated to be of silver, in later (2) recounting technique stated to be the same, electrodes again described as silver were qualified in an addendum as being of an alloy 'sterling silver'; in a later paper (6) in describing results of the second, the silver alloy electrodes were described as containing about 8 per cent of copper. *c*) The report of increased respiration in ground cerebral tissues or mitochondria derived from them (3) was not accompanied by any information on the effect of pulses in the absence of tissue nor on the behaviour of cell-containing tissues, in the same apparatus. It is noteworthy, however, that the oxygen uptake of liver mitochondria in the same experimental arrangement, was also reported as being increased by pulses.

Because of these differences in experimental arrangements, the detailed control experiments carried out with the apparatus showing no respiratory response in cell-free systems and reported in some eight categories (65, 66, 84), cannot be regarded as applying to those in which response was reported. These latter must therefore be

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regarded as not adequately controlled. They are subject also to the following experimental difficulties and uncertainties. *a*) The initially-reported stimulation of respiration of a cerebral suspension (3) was during experiments in which the control level of respiration was itself unstable, falling in absence of pulses to about 25 per cent of its initial level. With mitochondria, the total incubation time during which gas equilibration and measurement of respiratory change were recorded as being made was 10 minutes. Some of the uncertainties to which such experiments are subject have been recounted (96, 97). In addition, the difficulties of temperature equilibration of manometric apparatus in such brief periods would be accentuated by the thermal effect of the pulses applied (65). *b*) Pulses were later stated (2, 6) to appreciably increase respiration of cerebral mitochondria only when inorganic phosphate levels were below about 2 mM. However, the large increase initially reported (3) was in solutions 12 mM in phosphate. *c*) The large increase in respiration reported in liver and brain mitochondria (3) has since been stated not to occur with certain types of electrodes; pulses at silver-copper alloy electrodes were found to inhibit phosphorylation (6). They may thus also be detrimental in the cerebral system. *d*) The reported respiratory response was obtained with an electrode arrangement which enclosed only some 2 x 10 x 3 millimeters or 60 microliters, of a mitochondrial or ground tissue suspension 3 milliliters in volume. Field density appears certain to be much greater within the 60 microliters than beyond, and thus only $\frac{1}{50}$ of the suspensions were at any one time in the immediate fields of the applied pulses. This point is not discussed in the papers concerned, but appears to imply either a very much greater response by the mitochondria immediately within the electrodes, or a persistence of respiratory response in successive samples of the suspension after shaking has brought them within the electrodes; it could also, in absence of appropriate control experiments, be interpreted as an interaction of the charged electrodes with the fluid, possibly secondarily affecting the suspension.

Glucose and Glycolysis

In Situ. Determination of arterial and cerebral venous levels of a variety of substances in the blood of the cat, monkey, and man has shown the main substance normally oxidized by the brain to be glucose. Relevant data has recently been summarized (44, 73). The glucose is normally consumed at some 18 micromoles per gram of brain per hour, and largely converted to carbon dioxide but some 15 per cent of it appears as lactate and possibly $\frac{1}{10}$ of this amount as pyruvate. Blood analysis has also shown the glucose utilization and lactate formation by the brain to be altered markedly by agents which alter the level of cerebral activities. Each falls with general depressants, for example by some 35 per cent when Thiopental anesthesia reaches levels used surgically. Pentamethylenetetrazole, when inducing convulsions, increases glucose utilization and lactate formation. Data specifically concerning electrical excitation is meager and has not afforded values for the rate of change. Information has, however, been obtained by analysis not of the blood but of the brain itself.

Normal cerebral levels. The cerebral content of glucose and of metabolically related substances has been found capable of extremely rapid change with changed level of cerebral activities. This on the one hand provides a means of assessing the speed of chemical change in the brain; on the other hand it raises especial experimental difficulties in determining cerebral composition, for the metabolic changes can be considerable in a second or so. Methods capable of 'fixing' the brain rapidly *in situ* are required and the most successful and generally applicable of these has proved to

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be its freezing with liquid air or a similar agent (49, 50, 63, 73). The process was initially applied to anesthetized cats and dogs in which artificial respiration was maintained while the skull was opened so that at a chosen time, under chosen experimental conditions, the brain could be flooded with liquid air. Analysis thus gave the cerebral composition of the anesthetized animals (perhaps only lightly anesthetized) and the experimental conditions studied, for example hypoglycaemia or administration of convulsants, were superimposed on those of the anesthesia. Later workers (25, 91, 98) have in addition employed small experimental animals such as mice and young rats which can be dropped intact, with or without anesthesia or other treatment, into liquid air. Thermocouples show (91) that in 4–5 seconds the outer surface of the cerebral cortex of small rats treated in this way is frozen; between 5 and 15 seconds longer are needed for deeper parts of the brain to freeze. In assessing change of cerebral composition, these are relatively long times, as will be seen below; during them an unanesthetized animal can often be seen to struggle or otherwise react. When, however, the electroencephalogram was observed during the freezing, abnormal changes were not evident (87). The effects of the agents studied—depressants, excitants, or toxic substances—are nevertheless to some extent superimposed on those of the animal's reaction to the fixing process.

Acceptance of results obtained by these methods depends greatly on the following comparisons: *a*) between the two methods. In general, agreement is good, similar ranges of values being reported for several labile cerebral constituents of mice, rats, rabbits, cats and dogs, the first two species fixed intact and the others under light anesthesia and artificial respiration (73). Glucose was found at about 4 μM/gm, glycogen at 5 μM hexose/gm and lactate at 2 μM/gm. Hexose phosphates collectively are probably about 0.3 to 0.8 μM/gm (40, 58). *b*) With results when no special fixing methods are employed. These give very variable values for cerebral glycogen and phosphocreatine, all much lower than when tissue was fixed by rapid freezing (49, 50). Even tissues placed in liquid air promptly after decapitation show large accumulation of lactate (8). Similar removal of rat brain and placing it on solid CO_2 has been stated (5) to give the same results as use of liquid air *in vivo*, but as actual results quoted are extremely high (9 μM/gm) in hexose phosphates, this must be doubted. *c*) With results of other methods of fixing. Injection of iodoacetate, expected to inhibit lactate metabolism *in vivo*, gave values for cerebral lactate when the brain was later removed without freezing comparable to those obtained by freezing animals not treated with iodoacetate. *d*) Between animals variously treated before freezing. Effects of conditions judged to both increase and decrease cerebral activities, including apparently emotional excitement (91) are reflected in the cerebral composition of animals fixed by freezing; the effect of the fixing process itself is not, therefore, overwhelming. *e*) With the behavior of the isolated tissue, when conditions of fixing can be more rigidly controlled. Again, as will be indicated below, agreement is good.

The drastic process of freezing by liquid air thus appears to be the most satisfactory and thoroughly tested method available for determining labile cerebral constituents. In explanation of this, it has been pointed out (91) that thermocouple measurements show that in any one part of the brain, freezing is very sudden: successively deeper layers remain at 37°, possibly with some circulation remaining, until they suddenly fall to 0° C.; in animals dropped to liquid air, cold may block the transmission of peripheral sensations. Nevertheless, other procedures would be welcomed. Thus, it has been stated by a student of animal behavior that very rapid handling permits the mouse brain to be removed from the animal before fixation, and

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to yield glycogen values comparable with those of animals dropped to liquid air (18). Dexterity in handling the animals as well as the brain may be important in this. Other methods have been to sample the brain through a hole previously trepanned in the skull, with a scoop cooled in liquid air (82); or to inject an anesthetic immediately prior to decapitation (89).

Changes caused electrically. Electrically-induced changes in cerebral carbohydrates have been examined in cats, immobilized with a dihydro-β-erythroidine salt, and with the skull exposed and artificial respiration arranged under anesthesia (55, 86). When pulses (see below) were applied for 5 seconds across the skull, electrical recording from the brain showed the large potential changes characteristic of convulsive activity to begin some 8 seconds later. The convulsive discharge was allowed to run for 10 seconds the brain then fixed with liquid air and samples taken for analysis. Lactate then averaged 4.8 μM/gm in distinction to the value of 2.4 (S.E. 0.5) μM/gm of control animals. Supposing the increase to have taken place in the 23 seconds, its average rate would be 375 μM/gm/hr. Uncertainties in the time required for freezing and the period occupied by the change mean that the rate can best be described as between 250 and 600 μM/gm/hr. Similarly assessed, glucose was lost at 270 (180–400) μM/gm/hr. and glycogen lost at 310 (200–450) μM/gm/hr.

These rates are enormously greater than those of cerebral carbohydrate metabolism under normal conditions (see above) but in lactate and glucose have been paralleled during the action of other convulsive agents. Thus they can be induced in cats by bromocamphor, nicotinic acid diethylamide, and pentamethylenetetrazole (55) and in dogs by pentamethylenetetrazole (101). Change in glycogen has not proved consistent in direction in different experiments. In pentamethylenetetrazole convulsions in dogs somewhat differently prepared (34) no consistent change was found at different stages of convulsions even when glucose was lowered. Increase in glycogen was reported in mice during convulsions induced by several agents (16, 17). Here the change was found to be influenced by prior treatment of the mice with adrenaline or insulin and to be localized in the parts of the brain judged to be most affected by the convulsant.

The type of electrical stimulus employed to induce chemical change in the cat brain was chosen as one inducing convulsions, and was of brief square-wave pulses, peak potential 10 V, applied at 60 per second to electrodes 1 centimeter in diameter. In comparison with the therapeutic induction of electroshock in man with sine-wave currents of 60–100 V r.m.s. at electrodes 5–10 centimeters in diameter (43, 104) the conditions do not appear excessive. In man, the current flowing is some 750 milliamperes and only when this is increased 10-fold are momentary effects on cerebral blood vessels seen. This is relevant because the increase in lactate in the cat or dog might be thought due to anoxia. However, the increase persisted when the air respired was replaced by O_2-CO_2 (34, 100). Observations on cerebral cytochromes (73) make it unlikely that the oxygen utilization of the brain can be increased to more than double its normal value; during convulsions, the glucose utilized increased 15-fold. Thus increase in the normally anerobic reactions is to be ascribed the natural limit of the tissue's oxidative systems and not to lack of oxygen.

Perfused or Isolated Brain. Glucose utilization by the cat brain perfused with solutions simpler than whole blood (e.g., washed corpuscles suspended in albumin-glucose solutions) has been reported to be increased by pentamethylenetetrazole and by electrical pulses (26, 27). Such experimental arrangements are potentially very important, but to date have given results at variance both with those in the intact

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brain and in separated cerebral tissues. Thus the lactate of cerebral tissues of several species *in vivo* and *in vitro* has been found to be some 2 micromoles per gram; values found in the cat brain perfused with simplified solutions were 8–25 micromoles per gram. Increased glucose utilization *in vivo* and in separated tissues has been found within a few seconds of applied pulses, concomitantly with if not preceding increased respiration; in the brain perfused with simplified solutions, increased glucose utilization was found only subsequent to increased oxygen uptake (27). Respiratory studies in the isolated frog brain (see section on RESPIRATION) do not appear to have been paralleled by study of the substrate oxidized, but in the isolated frog spinal cord, electrical pulses led to increase in glucose utilization by 100–150 per cent and to increased output of carbon dioxide (106). Glycogen, it is interesting to note in comparison with the changes just described, increased after application of pulses.

Separated Tissues. *Maintained change in carbohydrate metabolism.* Under the circumstances in which applied pulses caused prolonged increase in cerebral respiration, they also increased glucose utilization and the formation of lactic acid (69, 75). Again, this is a property of cerebral and muscular and not other tissues (56). It occurs in all parts of the adult brain which have been examined, and in all species examined. In rat and guinea pig, the increase in lactic acid formation which was induced by pulses increased during development of the brain in much the same way as did the increment in respiration: again, therefore, paralleling other aspects of development (See section on RESPIRATION and ref. 33).

In oxygenated phosphate or bicarbonate salines with 10 mM glucose, the normal rate of glucose utilization by adult guinea pig cerebral cortex, of some 20 μM/gm/hr., was increased to 70 μM/gm/hr. and could remain at this level for over an hour while pulses were maintained. Oxidation of glucose accounts for only some 10 μM/gm/hr of this increase and the remainder has been accounted for almost entirely by increase in the formation of lactic acid (75). Thus a rate of lactate formation of 20 μM/gm/hr in absence of pulses, increased to 100 or 120 μM/gm/hr. within a minute of applying pulses; on stopping pulses, lactate formation again promptly reverted to its previous rate. Dependence of the increase on pulse-type has not been greatly studied but pulses of different characteristics minimal for maximal change in respiration bring about increase in lactate formation which again is or approaches maximal.

Effects of applied pulses on the metabolic fate of substrates other than glucose has been less completely explored. Of substances in whose presence pulses induce increased respiration (see above), 10 mM fructose with pulses yields also increased quantities of lactic acid. At lower levels it does not, presumably being preferentially oxidized; also, when no substrate is added to guinea pig cerebral tissues, the small quantity of lactate normally accumulating is decreased by pulses (69).

During the preparation of slices of cerebral cortex for metabolic experiments *in vitro*, the greater part of their glycogen is lost. It can, however, be caused to be resynthesized in oxygenated glucose salines (62, 81), but the rates of resynthesis observed have been slow, of 1–2 μM/gm/hr. Also the glycogen after resynthesis has proved relatively stable, being little affected by applied pulses during some minutes and while the pulses are bringing about large changes in glucose utilization or lactate formation. Such pulses do, however, prevent the resynthesis of glycogen if they are applied during the period when resynthesis would normally occur.

Transitory changes in carbohydrate metabolism. The maximum rate of glycolysis induced electrically in the experiments just described will be noted as involving the consumption of some 70 μM glucose/gm/hr., which is markedly lower than rates of

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glycolysis observed *in vivo* during brief periods of electrical excitation (see above). This difference has prompted the examination of the separated tissue also during periods much shorter than those just described, and of a few seconds only.

The experimental arrangement devised (41, 42, 81) supports sheets of tissue in a frame in a few milliliters of oxygenated saline. The frame carries electrodes and can be transferred from the saline to a fixing solution in 0.2 seconds, with pulses applied to the electrodes either throughout this period, or if necessary interrupted at defined periods before fixing. Penetration of fixing solutions to the thin sheet of tissue, kept extended between the electrodes, is rapid and enables the time-relations in fixing the separated tissue to be stated with much greater certainty than is possible in *in vivo* experiments (see above). The tissue in this apparatus metabolizes normally in the absence of pulses, and in their presence is fully capable of the prolonged changes in metabolism just described; but of more immediate interest are those taking place in a few seconds only. Prominent among these are a fall in glucose level and an increase in the lactate of the tissue.

During the first 20 seconds of pulses, these two changes proved to be stoichiometric: tissue glycogen, though present, changed little and 2 moles of lactate were formed for each mole of glucose lost. Moreover, the rate of change was in fact much greater than that of the sustained increase in glycolysis; the tissue's level of lactate, stable at about 1.5 μM/gm after an hour's metabolism, could be doubled in 12 seconds, corresponding to a rate of formation of some 400 μM/gm/hr. This rate is close to that judged to be occurring *in vivo* during 10 seconds of electrically-induced convulsive activity (see above). The commencement of lactate formation is prompt, within a second or two of pulses being applied; it is, however, transitory only and declines after 20 seconds, before the tissue's glucose is exhausted. It also exhibits an interesting inertia, in that, if pulses are stopped before the tissue's lactate is maximal, its formation continues for some seconds before declining. These phenomena reflect the linkage between metabolic responses to pulses and are discussed below.

Cell-Free Preparations. Although it has been stated that the oxidation of glucose by homogenates and mitochondria from the brain, was doubled by electrical pulses (4), this does not appear to be supported by glucose determinations or by evidence other than that previously discussed, where reasons were given for regarding it as doubtful. No alteration in lactic acid formation from glucose by cerebral suspensions has been found to be caused by applied pulses (84). It was earlier concluded (106) that destroying the cell structure of the frog spinal cord, though still permitting glucose utilization, abolished the increased utilization which normally followed application of electrical pulses.

Inorganic and Labile Phosphates

In Vivo. Analysis of arterial and cerebral venous blood has shown no appreciable change in level of inorganic phosphate on passage through the human brain (28); other phosphates appear to have been little examined in this respect. Isotopically labeled inorganic phosphate, intravenously in experimental animals, penetrates to the bulk of the brain only slowly though it enters more rapidly a few limited regions; from intracisternal injection exchange may be more rapid (10, 12, 24).

In contrast to the meager exchange of phosphate between bloodstream and brain, much of that in the brain itself is found to be undergoing extremely rapid change. Chemical methods for the fractionation and determination of the relevant phosphates in cerebral tissues are not discussed in detail in the present account, but it must be

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noted that many problems specific to the tissue arise in such determinations and that certain procedures devised for other tissues have proved inapplicable to those of the brain (38, 40, 58, 99a). This is especially important in working with isotopically labeled phosphates.

The rapid-fixation techniques found necessary to assess the cerebral content of labile substances were devised to a considerable extent in connection with determination of cerebral phosphates (49, 99). Whereas the brain fixed by ordinary methods contained large and variable quantities of inorganic phosphate, some 10–20 μM/gm, that adequately fixed contained only some 4 μM/gm, together with much greater quantities of substances which could yield inorganic phosphate with liberation of energy. Fixing the brain in both small and large experimental animals showed phosphocreatine at about 3.5 μM/gm and adenosine triphosphate at 2–3 μM/gm, whereas both were largely lost if fixation was not rapid. The criteria outlined in the section on GLUCOSE and GLYCOLYSIS judging the suitability of liquid air fixation for determination of carbohydrate constituents are applicable also in connection with the phosphates, and appropriate studies have shown the results reported below to be relevant to the brain *in vivo*. Thus, by fixing the brain of animals variously treated before fixation, defined changes have been found in cerebral phosphates (73); phosphocreatine increased by about one-third while inorganic phosphate decreased during the action of depressants (99); with various excitants these changes were reversed and adenosine triphosphate also fell.

Electrical pulses applied to the head of experimental animals for brief periods before fixation have afforded the best opportunity of measuring the speed at which the cerebral phosphates can be broken down and resynthesized *in vivo*. The rate of breakdown proved high. In cats, examined as described, phosphocreatine fell by 1.1 and acid-labile phosphates by 0.7 μM/gm during an experimental period of probably some 23 seconds; corresponding, therefore, to rates of some 170 and 110 μM/gm/hr., or within the limits of accuracy earlier discussed, a total loss of energy-rich phosphate of 200–400 μM/gm/hr. In young rats, suggestions of still more rapid change have been given (25). Here animals were immersed in liquid air, some without and others at fixed times after, applying current to electrodes connected to the scalp by a conducting jelly; 50 cycles per second sine-wave alternating current at 40 V was employed and resulted in relaxation followed by convulsions some 12 seconds after applying current. Electrical conditions may seem excessive; however, similar currents of some 5–100 milliamperes are used in rats in the experimental testing of anticonvulsants (104) and afford threshold values which vary with the physiological condition of the animals. In comparable experiments (91) some 5–15 seconds were required for freezing different depths of the animal's cerebral cortex, so that after applying current for 1 second, the change caused by it could have been in progress for 6–16 seconds before the tissue was fixed. In animals put to liquid air at this point, 1 second after current began, phosphocreatine had fallen from 3.1 to 1.35 μM/gm, corresponding to change at 400 to 1100 μM/gm/hr. Brief changes in adenosine triphosphate and inorganic phosphate also commenced within the same period, the observed rates being about half of that of the phosphocreatine.

In each of these groups of experiments it is not certain that maximum rates of change were observed, for the cerebral content of phosphocreatine did not undergo much greater change when pulses were applied for longer periods (60, or 2–5 sec.); briefer periods might therefore have afforded higher rates but were limited by the time required for freezing the tissue.

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Separated Tissues. The scope of studies with separated, cell-containing cerebral tissues was considerably increased when it was discovered (73, 76), that they could be caused to resynthesize and maintain defined levels of phosphocreatine and adenosine triphosphate. Thus the rapid loss of these substances which occurs when the tissues are prepared for *in vitro* studies is to a large extent reversible. Moreover, their *in vitro* levels proved susceptible of prompt response to a number of agents which affect the activities of the brain *in vivo*, including substrates, inhibitors, and stimulating agents. Also, added dinitrophenols and potassium salts at 20–100 mM rapidly depleted the tissue of the resynthesized labile phosphates (36, 67, 78). Of these various agents only electrical pulses will be assessed in detail here.

The effect of electrical pulses on the levels of inorganic, creatine and adenine phosphates in the separated tissue was found to be in the same sense as the changes just recounted *in vivo* (32, 37, 78). When measured as previously outlined in apparatus allowing the tissue to be fixed in a fraction of a second, the pulses acted remarkably promptly and induced the greatest metabolic rates yet measured in the intact tissue. Thus, for the first 1–2 seconds of applying pulses to guinea pig cerebral cortex, its adenosine triphosphate, initially at 2.2 μM/gm, fell at 600 μM/gm/hr. but then recovered its initial level as phosphocreatine, initially at 1.8 μM/gm, fell at 1500 μM/gm/hr. for some 3 seconds, reaching in this time about one-third its initial level. Inorganic phosphate increased by an amount equivalent to the fall in phosphocreatine, and at about two thirds the speed (37).

These changes could be brought about by pulses of characteristics about those minimal for maximum respiratory response by the tissue, including brief condenser pulses and also sine-wave alternating currents. Both the rate of change and the final level of phosphates varied with the type and intensity of applied current. Thus, to compare with the changes described *in vivo*, when sine-wave a.c. of 50 cycles per second and at 1.5 v r.m.s. was applied to tissue between electrodes 2 millimeters apart, phosphocreatine fell at 700 μM/gm/hr. from 1.5 to 0.9 μM/gm tissue (31). As the changes described were measured in tissues fixed at intervals of 1 or a few seconds, they afforded a clear picture of the speed and extent of the reactions studied. Maximal changes in phosphates occurred within 5 seconds, but the slower changes with less effective pulses could require 20 seconds; after maximal change tissue composition remained steady for some minutes while pulses continued at the same intensity: thus a new dynamic equilibrium had been established. If pulses were stopped shortly after the maximal change and incubation in oxygenated saline continued, labile phosphates were resynthesized at the expense of inorganic phosphate. Resynthesis proceeded steadily for 20 seconds at some 150 μM/gm/hr. until the original level was recovered (37).

Involvement of other constituents. Studies with isotopically-labeled phosphate have shown that in spite of the equivalence noted above between changes induced by pulses in inorganic, adenine and creatine phosphates, they are not self-contained but involve extensive exchange with other tissue constituents (39). In such experiments tissues were first allowed *in vitro* to resynthesize stable levels of the labile phosphates. A small amount of isotopic phosphate was then added; part of it rapidly appeared in each of the phosphate fractions examined, and with a suitably low level of external inorganic phosphates the isotopic content of the creatine phosphate and of the terminal phosphate of adenosine triphosphate was uniform in some 3 minutes. On transferring such a tissue to medium free from isotopically labeled phosphate, the labeled phosphate of the tissue's inorganic, adenine and creatine phosphates all

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gradually fell, partly on account of phosphate appearing in the external medium. The labeled phosphate was, however, also being distributed among other compounds of the tissue. Such transfer was greatly accelerated by a brief period (some 10 seconds) of applied electrical pulses. Pulses at this point brought about their usual fall in the level of creatine phosphate, but at the same time they lowered the isotopic content of the remaining phosphocreatine. Concomitantly, inorganic phosphate increased in actual amount but this also fell in isotopic content. Moreover, the labeled phosphate lost could be accounted for largely in other tissue constituents. Constituents extracted from the tissue by cold trichloracetic acid, when examined individually as in the above examples or collectively, all decreased in isotopic content, whether their actual amount rose or fell. Material not so extracted, which contained the greater part of the tissue's phosphate, changed little in total quantity of phosphate but became enriched in the isotope as a result of applied pulses. It will be recalled that the isotope was introduced during a period of 3 minutes incubation without pulses; nevertheless the isotopic content of the trichloroacetic acid-insoluble fractions was low and the 10 seconds of pulses increased it by some 30 per cent. Considering this value and the two time periods involved, it would appear that pulses bring about a very considerable increase in the rate of turnover of some as yet unidentified tissue constituents; thus events with pulses are more complex than simple breakdown of phosphocreatine even through adenosine triphosphate. Such complexity is necessary to any chemically feasible linkage of the labile phosphates to functional events in the tissue.

Cell-Free Preparations. As will be realized from the earlier description of respiration in cell-free cerebral preparations, the initial level and maintenance or accumulation of labile phosphates in cell-free preparations depends largely on the experimental choice of conditions. Thus adenosine triphosphate itself may be added (3, 13, 14, 73), with fluoride to retard its breakdown by adenosine triphosphatases which, initially latent, can become highly active in the ground tissue. An added agent can thus affect the labile phosphates of such mixtures by routes other than action on phosphorylation processes. For a given system and respiratory rate, the oxidative generation of labile phosphates has, however, a defined upper limit which with the pyruvate and dicarboxylic acid frequently chosen as substrates, is approximately 3 moles of phosphate per atom of oxygen absorbed. Many of the experimental arrangements chosen aim at maintaining relatively high P/O ratios, close to the value of 3, and this often involves careful and rapid handling of the ground tissue or of particles from it, conducting experiments at temperatures below 37°, and adding hexokinase and glucose to ensure that adenosine triphosphate generated is converted to a stable hexose phosphate, in which form it is indirectly measured. Alternatively, the disappearance of inorganic phosphate may be measured without direct measurement of a labile phosphate or phosphorylated product (2, 3, 13). Relationship between different subcellular particles in phosphorylation has not been as fully studied in cerebral as in other tissues, but mitochondrial fractions undoubtedly constitute the main site of oxidative phosphorylation.

Phosphorylation in cerebral suspensions and mitochondria has proved susceptible to certain agents which affect phosphorylation in other, more fully studied, mitochondrial preparations. This is notably so with 2:4-dinitrophenol at about 10^{-4}M, when P/O ratios can be decreased from 2.6 to a tenth of this value (15). Toxicity of the related 3:5-dinitrocresol when administered to rats in fact appears mainly due to this action at cerebral mitochondria (90). It will be recalled that 2:4-dinitrophenol, like increased concentrations of potassium salts and electrical pulses, decreased the

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labile phosphates of separated, cell-containing, cerebral tissues. However, studies with cerebral suspensions and mitochondria show that the modes of action of these agents are not alike. Whereas dinitrophenols affect the levels of labile phosphates in tissues by lowering their formation, attempts to show comparable phenomena with potassium salts in cerebral mitochondria revealed no such effect. Indeed, in studying cerebral and other mitochondria, concentrations of potassium salts in suspending fluids commonly imitate the intracellular potassium of tissue cells, which at some 100 mM is greater than that giving maximal depletion of the labile phosphates of the cell-containing tissue. Variation of potassium salts between 0.2 and 50 mM had little effect on oxidative phosphorylation by cerebral mitochondria (84).

The balance of evidence strongly indicates that this is also the case with applied electrical pulses. Thus, with several suspensions of guinea pig cerebral cortex, which esterified phosphate at 210 to 290 μM/gm tissue/hr., no significant change in phosphate esterification was found with applied pulses (84). The pulses were of types which, when applied in the same apparatus to finely-cut suspensions of cerebral tissues which were not ground but retained some cell-structure, caused maximal metabolic change. Also with mitochondrial preparations esterifying phosphate at 19–24 μM/mg of mitochondrial N per hour (about 300–350 μM phosphate per gram original tissue per hour), rates again showed little or no change under electrical conditions which greatly decreased the creatine phosphate content of the cell-containing tissue.

Results which have been concluded to indicate that such applied pulses decreased oxidative phosphorylation were reported under conditions stated as being, or being similar to, those in which pulses increased respiration of cell-containing cerebral tissues (3, 4). Actually the technique differed in the following ways, which appear relevant to the results quoted. *a*) Pulse types were different and no results were reported in their application to media in the absence of tissue preparations, nor to cell-containing tissues in the same apparatus. *b*) The electrodes were disposed to enclose only some 2 per cent of the reaction mixture and were apparently of a copper-silver alloy. Pulses applied to them were reported to decrease phosphorylation not only by cerebral preparations but also by those from kidney (3). In the latter preparations they were thus inducing a change for which no parallel was available in the cell-containing tissue. Phosphorylation by cerebral mitochondria has been found (19) extremely susceptible to many agents including silver ions, being inhibited by 1–10 μM silver salts and also by the mere presence of metallic silver in reaction mixtures. Pulses can also, probably by various mechanisms inhibit isolated enzyme preparations (84). Thus influences from which the cell-containing tissue is immune have major but from the present point of view extraneous effects in cell-free systems. Many of these are irreversible; reversibility has been claimed for the effect of pulses on phosphorylation by mitochondria but experiments reported (2) are not convincing. Pulses have also been concluded to decrease phosphorylation from measurements of the specific activity of phosphates esterified after adding labeled inorganic phosphate (3). As activities of other phosphates were not measured and silver-copper electrodes were apparently used, considerable scope remains for other explanations (see above, and ref. 39). It is conceivable that in exciting intact cerebral tissues, metabolically derived energy might be diverted to perform additional osmotic work rather than appear as phosphate bond energy (73), but this lacks direct evidence.

Interconnections and Conclusions

Investigations of the brain *in situ*, of fragments of it in isolation, and of cell-free preparations from it are thus on the whole consistent and give complementary and

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characteristically different types of information. Information from systems of each sort has been necessary to show how the level of cerebral metabolism may be adjusted to a changed level of cerebral activity. This subject has been assessed recently (73) and will only incidentally be recounted here; instead, attention will be directed to problems and opportunities which arise when information from the different systems is compared.

Cerebral Tissues in Situ and in Isolation. It is now evident that mammalian cerebral tissues are much more autonomous than was believed a decade ago, when measurement of their respiratory rate was described as valuable but ". . . hardly expected to approach the actual values for the intact living brain because of the damage to the tissue from the mincing and slicing, the anoxia incurred during the preparation and the artificiality of the medium in which the tissue is placed" (35). It is especially impressive to find that the separated tissue, in spite of the damage involved in replacing the capillary system by a series of cuts at 0.4 millimeter intervals, resynthesized its pattern of labile metabolites. Conceivably, resynthesis is to only about two-thirds of the *in vivo* levels because of the cellular damage. This is one of the quantitative differences remaining between the tissue *in vivo* and *in vitro*; either respiration is lower *in vitro*, or, when stimulated by pulses, it may reach *in vivo* levels but the tissue is then of lower phosphocreatine content than obtains *in vivo*. However, the devising of experimental fluids specifically in order to maintain the composition of the separated tissue as close as possible to that *in vivo* has as yet hardly begun.

Still more encouraging has been the finding of comparable metabolic responses *in vivo* and *in vitro* to electrical pulses. Comparison *in vivo* was made with pulses which produced the drastic result of a convulsion. It has been urged that convulsive activity has its counterpart in natural aspects of animal behavior; certain of the chemical changes so induced occur also in excitement which is largely emotional (51*a*, 91). From a chemical point of view, it is understandable that investigators commencing the study of the material changes associated with different levels of cerebral activity should have chosen conditions which brought as much as possible of the brain to a greater or lesser degree of activity, in order that analysis of the organ should show appreciable changes. It may perhaps be assumed, unless proven otherwise, that the changes in oxygen or phosphates which occur in the whole brain when it is stimulated to such a degree that it all reacts are comparable to those which occur normally in its different parts as they come successively into greater activity. Against the background of similar response in oxygen, lactate, glucose and phosphates when pulses are applied *in vivo* and *in vitro*, may be noted phenomena in one system whose counterpart in the other is uncertain or lacking. The glycogen changes reported *in vivo* have not been reproduced *in vitro*; those in acetylcholine and ammonia have been in part only (93) but here *in vitro* observations have not yet employed sufficiently brief periods of time. For other comparisons, see (73).

Tissues and Cell-Free Preparations in Vitro. It is a reflection of the cell as basic biological unit that several phenomena traceable from brain to cerebral slice are lost or transformed in cell-free preparations. Noting first those which remain, maximal respiratory rate, glycolytic rate, and the connection of these with phosphorylation, afford accurate quantitative comparisons. The ways in which the maxima are achieved, however, differ notably. Inorganic phosphate and phosphate acceptors appear in the cell to be part of a delicate and ingenious self-regulating mechanism (see ref. 73). Energy utilization produces them, they stimulate the energy-yielding processes in the course of which they are consumed with formation of, again, the reserves of energy-rich substances of the cell; with accumulation of these substances

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and fall in phosphate level, respiration and glycolysis revert to their former, unstimulated levels. Cerebral tissues respiring at 60–120 μM O_2/gm/hr. could at P/O ratios of 2.5 to 3, esterify 300 to 700 μM phosphate/gm/hr. and thus consume the 3 or 4 micromoles of phosphate found per gram of tissue in 15–40 seconds. Thus, unless inorganic phosphate is continuously formed or the coupling of phosphorylation with respiration released, respiration would be limited to this period. Actually, in the anesthetized brain *in vivo* or in unstimulated tissue *in vitro*, respiration continues at some 60 μM/gm/hr. Under these conditions, though there may be few if any signs of transmitted electrical impulses from the cells, standing potentials and large differences in concentrations of diffusible substances are maintained across the very considerable area of fine, partly permeable cell walls. This and other aspects of cell maintenance require metabolically derived energy presumably equivalent to the 60 μM O_2/gm/hr.

Respiration in otherwise adequate cell-free systems is increased simply by providing more inorganic phosphate or phosphate acceptors. Providing these to phosphate-deficient media in which cell-containing tissues are suspended has no comparable effect. Analysis of the tissue, however, shows that agents which increase its respiration increase its native inorganic phosphate. This is achieved in the case of dinitrophenols by their preventing the phosphorylation usually associated with respiration. Such an explanation is quantitatively inadequate in the case of electrical pulses applied to the cell-containing tissue, for these cause loss of labile phosphates at 500 μM/gm/hr. Respiration has been observed to allow accumulation of phosphates at only 150 μM/gm/hr.; the maximum provided was noted above to be 300–700 MM/gm/hr. On the other hand, the tissue possesses potent systems transferring or degrading the labile phosphates at rates fully adequate to form a basis for loss at the greatest rates observed: creatine phosphokinase at 3750 μM/gm/hr. (83) and adenosine triphosphatase at 2000 μM/gm/hr. (29).

Cell-Free Systems and Individual Enzymes. It is intriguing at the present state of knowledge to query whether the systems known to act on the labile phosphates at adequate rate actually constitute the route of their loss when pulses or analogous agents induce it. There appears no difficulty in accepting creatine phosphokinase as mediating in the loss of phosphocreatine by transfer to adenosine diphosphate. It operates at the high rate quoted at the pH and the concentration of phosphocreatine native to the cell (83) and apparently at low adenosine diphosphate levels. On the other hand, adenosine triphosphatase as observed when adenosine triphosphate is added to cerebral extracts is inadequate as basis for the changes observed in adenosine triphosphate in the cell-containing tissue: for the inorganic phosphate comes from another source. As, however, certain phosphatases have been shown capable of phosphate transfer between organic substrates, it is feasible that the adenosine triphosphatase of the brain, as of muscle, transfers phosphate of adenosine triphosphate to another tissue-constituent, at present characterized only as being trichloroacetic-acid insoluble. How then does this constituent become available as acceptor by a few electrical pulses in a second or so? An economical hypothesis would be that it had lost the inorganic phosphate which newly appears with pulses (unlabeled in the experiments described in the section on INORGANIC AND LABILE PHOSPHATES) and is then in its turn rephosphorylated. Adenosine triphosphate generation is mainly at mitochondria; much of the formed triphosphate is free in the cytoplasm; ion movements likely to represent the tissue's energy-consuming processes take place at the cell membrane. An agent such as adenosine triphosphate appears necessary between the mitochondria and cell-membrane.

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These observations bear on the type of phenomena which in cell-free systems can be expected to be the analogue or basis of metabolic response to pulses in cell-containing ones. Reasons have been given above for considering that changes reported in oxidation and phosphorylation in cerebral suspensions and mitochondria are inadequate in this respect. On the other hand, interaction has been found between effects of pulses and of protoveratrine (107); and protoveratrine and pulses have, in cell-containing tissues, effects reminiscent of certain found in mitochondria (46, 74). Individual enzymes exposed to electrical currents or pulses can be activated or inactivated through oxidation or ionic effects (see ref. 84) but to date have not offered a basis for the stimulation in cell-containing tissues. Presumably the pulses allow ion migration at the cell-membrane, in the direction of existing ion gradients and too rapid to be enzymic: action possibly not enzymic at a cell-membrane constituent is thus necessary. The ion migration is reversed by the energy-yielding systems through adenosine triphosphate: tracing the fate of the transferred phosphate may thus characterize a component involved in performing the tissue's osmotic work.

In this and in a preceding survey (73) it will be noted that the theme of the speed of chemical change in the brain has been developed more extensively than hitherto; it has been applied in tracing interconnections between phenomena of disparate chemical categories and at different levels of cellular and tissue organization, and in excluding or showing the feasibility of specific biochemical mechanisms. To Sherrington's eloquent description of the pattern of electrical change in the brain is being gradually added a much more intricate pattern of chemical change. For this in a single cell is highly localized in space to specific parts of subcellular components, and moreover is highly differentiated in time to events of very different speed before, during, and after a given cell discharge.

REFERENCES

1. ABOOD, L. G. In: *Phosphorus Metabolism.* Baltimore: John Hopkins Press, 1952, vol. 2, p. 822.
2. ABOOD, L. G. *Am. J. Physiol.* 176: 247–252, 1954.
3. ABOOD, L. G., R. W. GERARD AND S. OCHS. *Am. J. Physiol.* 171: 134–139, 1952.
4. ABOOD, L. G., R. W. GERARD AND R. D. TSCHIRGI. In: *Phosphorus Metabolism.* Baltimore: John Hopkins Press, 1952, vol. 2, p. 798.
5. ABOOD, L. G., E. KUN AND E. M. K. GEILING. *J. Pharmacol. & Exper. Therap.* 98: 373–379, 1950.
6. ABOOD, L. G. AND L. ROMANCHEK. *Biochem. J.* 60: 233–238, 1955.
7. ADRIAN, E. D. AND F. J. J. BUYTENDYK. *J. Physiol.* 71: 121–135, 1931.
8. AVERY, B. F., S. E. KERR AND M. GHANTUS. *J. Biol. Chem.* 110: 637–642, 1935.
9. AYRES, P. J. W. AND H. MCILWAIN. *Biochem. J.* 55: 607–617, 1953.
10. BACKAY, L. *Arch. Neurol. & Psychiat.* 71: 673–683, 1954.
11. BANGA, I., S. OCHOA AND R. A. PETERS. *Biochem. J.* 33: 1980–1996, 1939.
12. BORELL, U. AND A. ÖRSTROM. *Biochem. J.* 41: 398–403, 1947.
13. BRODY, T. M. AND J. A. BAIN. *J. Biol. Chem.* 195: 685–696, 1952.
14. BRODY, T. M., R. I. H. WANG AND J. A. BAIN. *J. Biol. Chem.* 198: 821–826, 1952.
15. CASE, E. M. AND H. MCILWAIN. *Biochem. J.* 48: 1–11, 1951.
16. CHANCE, M. R. A. *Brit. J. Pharmacol.* 6: 1–7, 1951.
17. CHANCE, M. R. A. AND A. E. WALKER. *J. Endocrinol.* 10: 316–324, 1954.
18. CHANCE, M. R. A. AND D. C. YAXLEY. *J. Exper. Biol.* 27: 311–323, 1950.
19. CHAPPELL, J. B. AND G. D. GREVILLE. *Nature, London* 174: 930–931, 1954.
20. CHUTE, A. L. AND D. H. SYMTH. *Quart. J. Exper. Physiol.* 29: 379–394, 1939.
21. CONNELLY, C. M., D. W. BRONK AND F. BRINK. *Rev. Sci. Instr.* 24: 683–695, 1953.
22. CRAIN, S. M. *Proc. Soc. Exper. Biol. & Med.* 81: 49–51, 1952.
23. DAVIS, E. W., W. S. MCCULLOCH AND E. ROSEMAN. *Am. J. Psychiat.* 100: 825–829, 1944.
24. DAWSON, R. M. C. *Biochem. Soc. Symp.* 8: 93–102, 1952.
25. DAWSON, R. M. C. AND D. RICHTER. *Am. J. Physiol.* 160: 203–211, 1950.

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26. GEIGER, A. J. MAGNES, R. M. TAYLOR AND M. VERALLI. *Am. J. Physiol.* 177: 138–149, 1954.
27. GERARD, R. W. *Science* 122: 225–230, 1955.
28. GIBBS, E. L., W. G. LENNOX., L. F. NIMS AND F. A. GIBBS. *J. Biol. Chem.* 144: 325–332, 1942.
29. GORE, M. B. R. *Biochem. J.* 50: 18–24, 1951.
30. GORE, M. B. R. AND H. MCILWAIN. *J. Physiol.* 117: 471–483, 1952.
31. GREENGARD, O. *Thesis*: University of London, 1955.
32. GREENGARD, O. AND H. MCILWAIN. *Biochem. J.* 61: 61–68, 1955, and unpublished observations.
33. GREENGARD, P. AND H. MCILWAIN. *Proc. 1st Internat. Neurochem. Symp.* New York: Acad. Press, 1955, p. 251–260.
34. GURDJIAN, E. S., J. E. WEBSTER AND W. E. STONE. *Proc. A. Res. Nerv. & Ment. Dis.* 26: 184–204, 1946.
35. HANDLEY, C. A., H. M. SWEENEY, Q. SCHERMAN AND R. SEVERANCE. *Am. J. Physiol.* 140: 190–196, 1943.
36. HEALD, P. J. *Biochem. J.* 55: 625–631, 1953.
37. HEALD, P. J. *Biochem. J.* 57: 673–679, 1954.
38. HEALD, P. J. *Biochem. J.* 63: 235–242, 1956.
39. HEALD, P. J. *Biochem. J.* 63: 242–250, 1956.
40. HEALD, P. J. *Maudsley Monograph, 1956*. In preparation.
41. HEALD, P. J. AND H. MCILWAIN. *Biochem. J.* 63: 231–235, 1956.
42. HEALD, P. J., H. MCILWAIN AND D. NELLIGAN. *Biochem. J.* 56: XXXIX, 1954.
43. HEMPHILL, R. E. AND W. GREY-WALTER. *J. Ment. Sc.* 87: 256–275, 1941.
44. HIMWICH, H. E. *Brain Metabolism and Cerebral Disorders*. Baltimore: Williams & Wilkins, 1951.
45. HODGKIN, A. L. AND B. KATZ. *J. Physiol.* 108: 37–77, 1949.
46. HUNTER, F. E., JR. AND L. FORD. *J. Biol. Chem.* 216: 357–369, 1955.
47. JASPER, H. H., C. S. BRIDGMAN AND L. CARMICHAEL. *J. Exper. Psychol.* 21: 63–71, 1937.
48. KATZ, B. *Electric Excitation of Nerve*. Oxford: University Press, 1939.
49. KERR, S. E. *J. Biol. Chem.* 110: 625–635, 1935.
50. KERR, S. E. *J. Biol. Chem.* 116: 1–7, 1936.
51. KETY, S. S. In: *Methods in Medical Research*. Chicago: Yr. Bk. Pub., 1948, vol. 1, pp. 204–216.
51a. KETY, S. S. *Neuropharmacology* 1: 13–83, 1955.
52. KETY, S. S. AND C. F. SCHMIDT. *J. Clin. Invest.* 27: 476–483, 1948.
53. KETY, S. S., R. B. WOODFORD, M. H. HARMEL, F. A. FREYHAN, K. E. APPEL AND C. F. SCHMIDT. *Am. J. Psychiat.* 104: 765–770, 1948.
54. KIMEL, V. M. AND F. KAVALER. *J. Comp. Neurol.* 94: 257–265, 1951.
55. KLEIN, J. R. AND N. S. OLSEN. *J. Biol. Chem.* 167: 747–756, 1947.
56. KRATZING, C. C. *Biochem. J.* 50: 253–257, 1951.
57. KRATZING, C. C. *Biochem. J.* 54: 312–317, 1953.
58. KRATZING, C. C. AND A. NARAYANASWAMI. *Biochem. J.* 54: 317–323, 1953.
59. KREBS, H. A. *Biochim. et biophys. acta* 4: 249–269, 1950.
60. KREBS, H. A. AND K. HENSELEIT. *Ztsclr. f. physiol. Chem.* 210: 33–66, 1932.
61. LARRABEE, M. G. AND D. W. BRONK. *Cold Spring Harbour Symp. Quant. Biol.* 17: 245–266, 1952.
62. LEBARON, F. N. *Biochem. J.* 61: 80–85, 1955.
63. LEPAGE, G. A. In: *Methods in Medical Research*. Chicago: Yr. Bk. Pub. 1948, vol. 1, pp. 337–357.
64. LIBET, B. AND R. W. GERARD. *J. Neurophysiol.* 4: 438–455, 1941.
65. MCILWAIN, H. *Biochem. J.* 49: 382–393, 1951.
66. MCILWAIN, H. *Biochem. J.* 50: 132–140, 1951.
67. MCILWAIN, H. *Biochem. J.* 52: 289–295, 1952.
68. MCILWAIN, H. *Biochem. Soc. Symp.* 8: 27–43, 1952.
69. MCILWAIN, H. *Biochem. J.* 55: 618–624, 1953.
70. MCILWAIN, H. *J. Neurol. Neurosurg. & Psychiat.* 16: 257–266, 1953.
71. MCILWAIN, H. *Arch. Neurol. Psychiat.* 71: 488–495, 1954.
72. MCILWAIN, H. *J. Physiol.* 124: 117–129, 1954.
73. MCILWAIN, H. *Biochemistry and the Central Nervous System*. London: Churchill, 1955.
74. MCILWAIN, H. *Biochem. J.* 63: 257–264, 1956.
75. MCILWAIN, H., G. ANGUIANO AND J. D. CHESHIRE. *Biochem. J.* 50: 12–18, 1951.
76. MCILWAIN, H., L. BUCHEL AND J. D. CHESHIRE. *Biochem. J.* 48: 12–20, 1951.
77. MCILWAIN, H. AND H. L. BUDDLE. *Biochem. J.* 53: 412–420, 1953.
78. MCILWAIN, H. AND M. B. R. GORE. *Biochem. J.* 50: 24–28, 1951.

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79. McIlwain, H. and M. B. R. Gore. *Biochem. J.* 54: 305–312, 1953.
80. McIlwain, H., S. Ochs and R. W. Gerard. *Am. J. Physiol.* 171: 128–133, 1952.
81. McIlwain, H. and M. Tresize. *Biochem. J.* In press.
82. Minaev, P. F. and T. P. Kurokhtiva. *Ukrain. Biokim. Zhur.* 21: 359–362, 1949.
83. Narayanaswami, A. *Biochem. J.* 52: 295–301, 1952.
84. Narayanaswami, A. and H. McIlwain. *Biochem. J.* 57: 663–666, 1954.
85. Ochoa, S. *J. Biol. Chem.* 141: 245–251, 1941.
86. Olsen, N. S. and J. R. Klein. *Proc. A. Res. Nerv. & Ment. Dis.* 26: 118–130, 1946.
87. Olsen, N. S. and J. R. Klein. *J. Biol. Chem.* 167: 739–756, 1947.
88. Overton, E. *Pflügers Arch. ges. Physiol.* 92: 346–386, 1902.
89. Palladin, A. V. *Zhur. Vysshei. Nervnoi. Deyatel I.P. Pavlova* 3: 801–808, 1953.
90. Parker, V. H. *Biochem. J.* 57: 381–386, 1954.
91. Richter, D. and R. M. C. Dawson. *Am. J. Physiol.* 154: 73–79, 1948.
92. Rodnight, R. and H. McIlwain. *Biochem. J.* 57: 649–661, 1954.
93. Rowsell, E. V. *Biochem. J.* 57: 666–673, 1954.
94. Ruf, H. *Arch. f. Psychiatrie. Nervenkrankh.* 187: 97–127, 1951.
95. Schmidt, C. F., S. S. Kety and H. H. Pennes. *Am. J. Physiol.* 143: 33–52, 1945.
96. Slater, E. C. and F. A. Holton. *Biochem. J.* 55: 530–544, 1953.
97. Slater, E. C. and F. A. Holton. *Biochem. J.* 56: 28–40, 1954.
98. Stone, W. E. *Biochem. J.* 32: 1908–1918, 1938.
99. Stone, W. E. *J. Biol. Chem.* 135: 43–50, 1940.
99a. Stone, W. E. In: *Methods in Medical Research.* Chicago: Yr. Bk. Pub., 1948, vol. 1, pp. 353–357.
100. Stone, W. E. In: *Biology of Mental Health and Disease:* New York: Hoeber, 1952, p. 564.
101. Stone, W. E., J. E. Webster and E. S. Gurdjian. *J. Neurophysiol.* 8: 233–350, 1945.
102. Sutherland, V. C., T. N. Burbridge and H. W. Elliott. *Am. J. Physiol.* 180: 195–201, 1955.
103. Thomas, J. and H. McIlwain. Submitted for publication.
104. Toman, J. E. P., E. A. Swinyard, M. Merkin and L. S. Goodman. *J. Neuropath. & Exper. Neurol.* 7: 35–46, 1948.
105. Weil-Malherbe, H. *Biochem. J.* 32: 2257–2275, 1938.
106. Winterstein, H. In: *Handbuch der Normalen und Pathologischen Physiologie.* Berlin: Springer, 1929, pp. 515–611.
107. Wollenberger, A. *Biochem. J.* 61: 77–80, 1955.

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